

POTPRESS · DESIGN PRECIS

PotPress design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

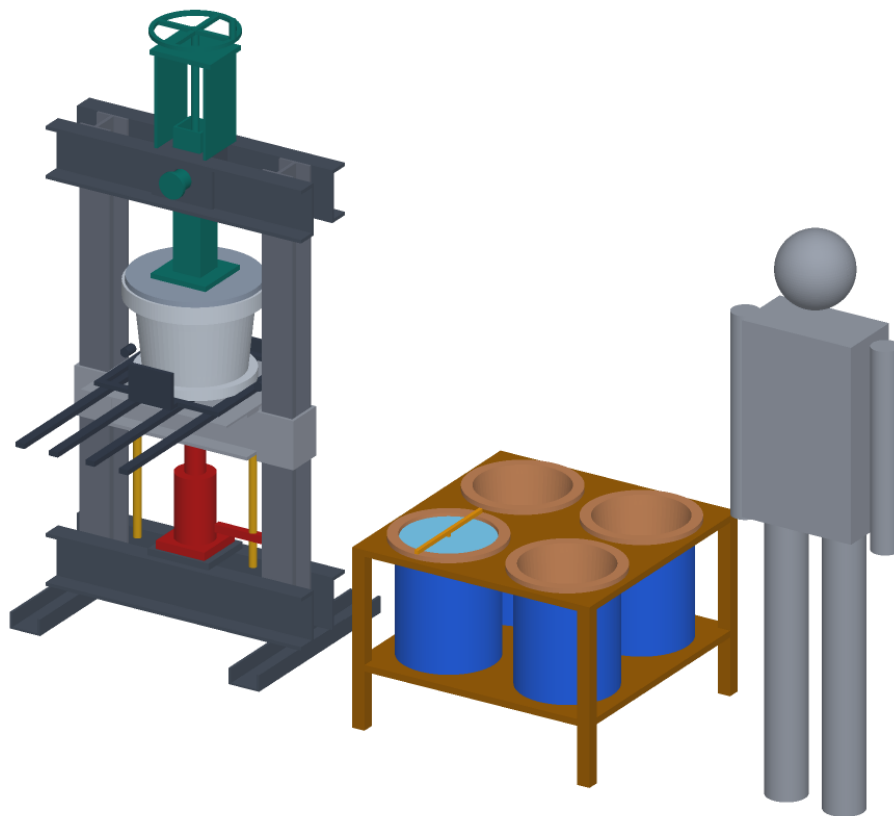
Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update; design choices decided by Amish (PPR-DDR-001); numbers corrected to PPR-CAL-001 (shell molds, 2 x UPN 160 base, one 60 mm pin, 2 x 2 QC rack, cost)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, material flow, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

PotPress design precis

PotPress is a hand-pumped hydraulic press that forms one ceramic pot filter per stroke between a cast aluminum female mold and male mold, plus a four-station rack for the standard one-hour flow-rate test. A 20 t bottle jack on the base lifts a guided platen carrying the female mold against a fixed male mold, whose stem is held by one load pin in the top crossbeam for pressing and raised by a hand crank for loading. One parametric filter geometry drives the molds, the 3D-printed casting patterns and the printed flow gauge. The TRL 3 calculations (PPR-CAL-001) give a cycle of about 5.2 min and about 69 pots per 6 h, a strong frame at 1.5 times the jack rating, and a parts cost of \$926, which is over the \$720 budget. The design choices below were decided by Amish on 2026-09-25 (PPR-DDR-001).

PotPress

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale

Figure 1. PotPress with the 2 x 2 QC flow-test rack and a 1.75 m person for scale. The press is shown closed at the end of a pressing stroke.

■ How it works

1. **Load.** With the platen down, the female mold rim is at about 921 mm. The operator slides the carriage out 430 mm on its rails, lays a polyethylene release liner in the female mold, places a weighed charge of about 7.5 kg of clay, burnout and water mix, and slides the carriage back to its end stop.

2. **Lower the male mold.** A handwheel on top of the frame turns a self-locking Tr24 x 5 lead screw that runs the male mold stem down 200 mm (40 turns) through the top crossbeam until the pin bores line up. The operator inserts the 60 mm load pin through the beam and stem. From then on the pin, not the screw, carries the press force.
3. **Press.** The operator closes the front gate and pumps the jack, about 70 strokes. The platen, guided by sleeves on both uprights, lifts the female mold about 110 mm and squeezes the charge into the gap between the molds. The male flange centers in a tapered lip on the female mold and lands on a stop face, so the 15 mm wall and rim are set by the molds, not by how hard the operator pumps. Excess mix squeezes into a flash groove at the rim for trimming.
4. **Open and demold.** The operator opens the jack release; the platen's weight and two springs return it. The operator pulls the pin and cranks the male mold up, leaving its tip 70 mm above the female rim. The carriage slides out toward the operator and tilts on its pivots so the pot, still in its liner, turns out onto a drying board.
5. **Dry, fire, treat.** Drying, firing and silver treatment follow the factory's existing practice and are outside PotPress.
6. **Test.** Fired pots are soaked, hung by their rims in the QC rack over collection buckets, filled to 10 mm below the rim and read after one hour with the printed T-gauge. The reading is corrected to 25 °C with the water temperature.

PotPress: cutaway

CONCEPT, NOT FOR FABRICATION

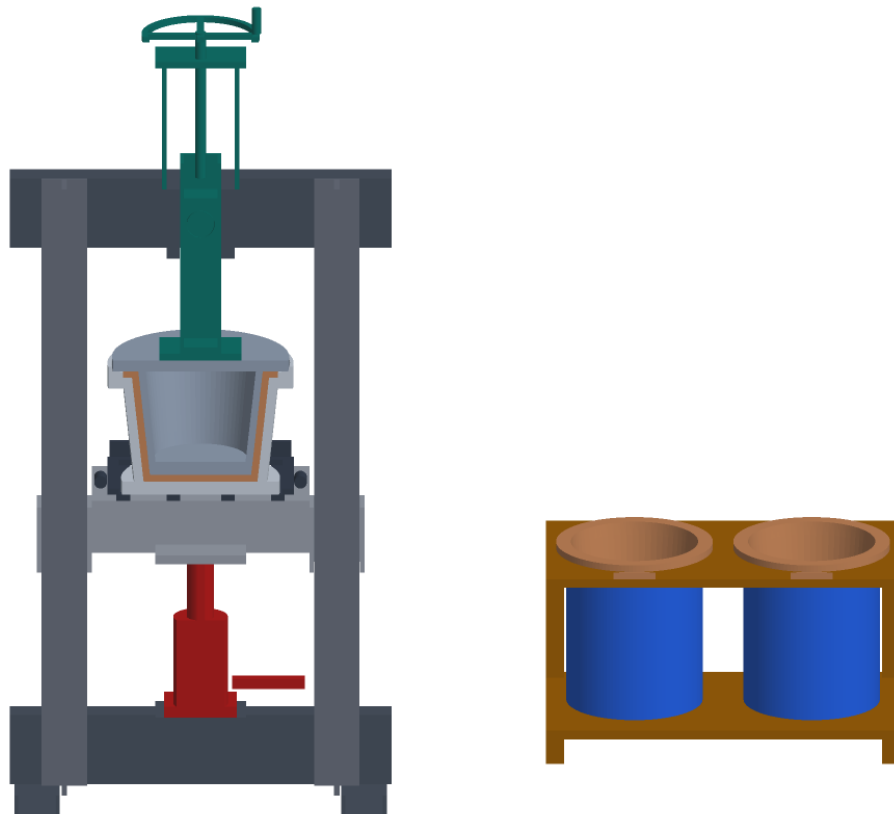


Figure 2. Cutaway through the closed press and the QC rack, showing the pressed pot between the molds, the shell molds and the load path.

PotPress: material flow per filter, mixed charge to passed filter (all values are estimates)

CONCEPT, NOT FOR FABRICATION

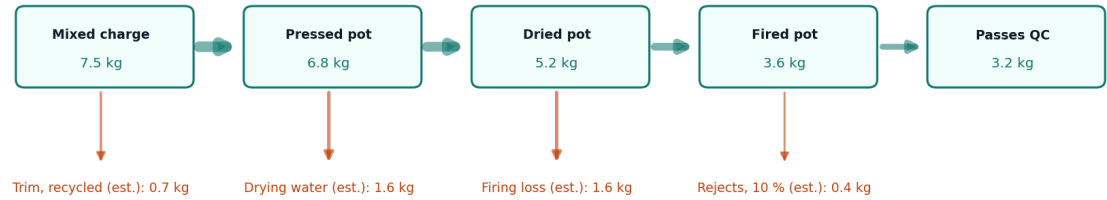


Figure 3. Material flow for one filter, from mixed charge to a filter that passes QC, in kg. All values are estimates from PPR-CAL-001.

■ Main components

Numbers match the exploded view (Figure 4), `bom/bom.csv` and the general arrangement drawing PPR-DWG-001.

Table 1. Main components.

#	Component	Choice	Notes
1	Base beam and feet	Two UPN 160 channels 840 mm with a 100 mm gap, 20 mm jack plate, two UPN 100 feet bolted on	Same section as the top beam; the TRL 2 single UPN 100 would reach about 1,071 MPa
2	Uprights	Two UPN 100 channels back to back each side, about 1.40 m, sandwiched in the beam gaps at 600 mm centers	54 MPa in tension at 294 kN
3	Top crossbeam	Two UPN 160 channels, 12 mm web doublers and a 62 mm pin bore at the center, UHMW-PE stem guides	190 MPa at 294 kN
4	20 t bottle jack	Standard jack, 150 mm stroke, about 60 mm ram	Uses 110 mm of stroke
5	Moving platen	Two UPN 140 channels, 6 mm deck, jack pad, 150 mm guide sleeves on the uprights	38.9 kg, the heaviest part as fabricated
6	Return springs	Two tension springs	Gravity returns the platen; the springs keep it seated
7	Rails and carriage	Four flat-bar rails at 60 and 170 mm each side, extending 430 mm to the front; carriage frame with handle and tilt pivots	Moves the female mold out of the pinch zone
8	Female mold	Cast aluminum shell, 20 mm behind the cavity, 30 mm solid base, 410 mm flange with a locating lip, rim counterbore and flash groove; 24.3 kg	A solid block would weigh about 57 kg
9	Male mold	Cast aluminum hollow plug, 15 mm shell and 30 mm tip, 390 mm rim-forming flange; 17.9 kg	A solid plug would weigh about 41 kg
10	Male mold slide	SHS 90 x 8 stem with a pin block, one 60 mm 42CrMo4 load pin, crank bracket, Tr24 x 5 lead screw and 260 mm handwheel	The screw only lifts; the pin carries the load
11	Pressed filter pot	The product, shown for context	Not costed
12	QC flow-test rack	2 x 2 stations, 780 x 780 x 550 mm, steel angle and plywood	Four rims cannot fit in R11's 0.8 x 0.5 m
13	Printed T-gauge	PETG dipstick; 1.0 L is 16.7 mm on the scale	Scale generated from the geometry
14	Collection buckets	20 L food-grade HDPE	

Casting patterns, guards and front gate with interlocks, release liners, fasteners and consumables, and the QC timer and thermometer are BOM items 15 to 19 and are not modeled.

PotPress: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Base beam, feet and jack plate
- 2 Uprights
- 3 Top crossbeam with pin doublers
- 4 20 t bottle jack
- 5 Moving platen with guide sleeves
- 6 Return springs
- 7 Rails and mold carriage
- 8 Female mold (cast aluminum)
- 9 Male mold (cast aluminum)
- 10 Male mold slide, load pin, crank
- 11 Pressed filter pot
- 12 QC flow-test rack (2 x 2)
- 13 Printed T-gauge
- 14 Collection buckets

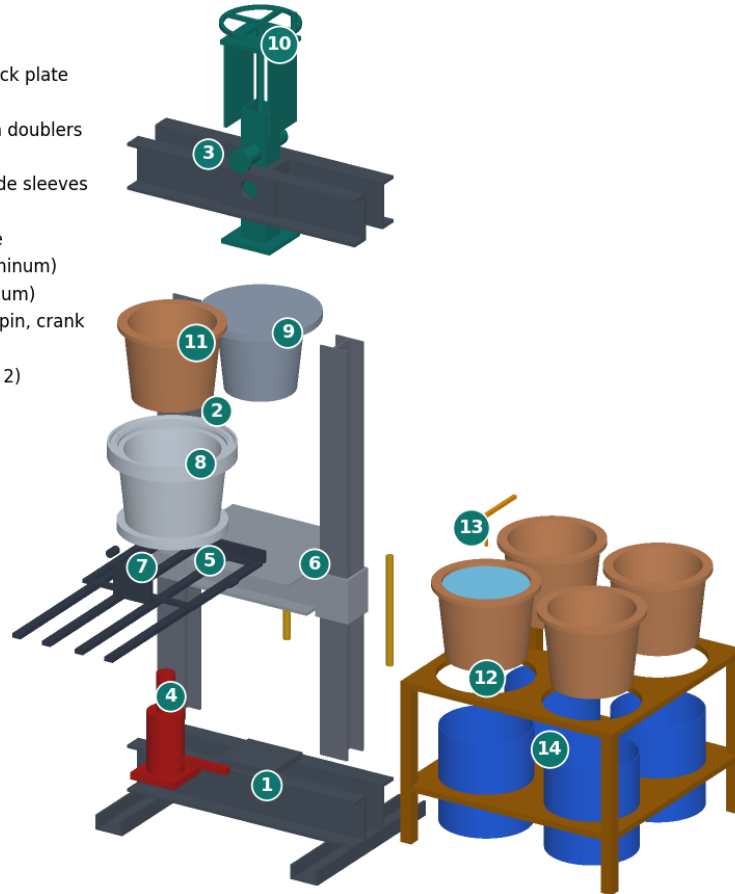


Figure 4. Exploded view with numbered callouts matching the BOM.

■ Key numbers

All values are estimates from PPR-CAL-001, which lists its assumptions and the results against every requirement.

Table 2. Key numbers at TRL 3.

Quantity	Value	Basis
Filter volume	12.30 L to the brim; 9.91 L working	Inner rim 280 mm, floor 230 mm, depth 240 mm
Pot masses	Charge 7.47 kg; pressed 6.79 kg; fired 3.61 kg	RDI-C recipe, 1.64 kg/L pressed
Mean pressure	0.52 to 1.05 MPa at 5 to 10 t; 2.10 MPa at 20 t	0.0935 m ² projected
Frame at 294 kN	Beams 190 MPa, uprights 54 MPa, pin 258 MPa (yield 650), welds 62 MPa	S275; 42CrMo4 pin
Deflection between molds	0.61 mm at 10 t; 1.84 mm at 294 kN	R3 (1 mm at 294 kN) not met
Travel	110 mm pressing of 150 mm stroke; 310 mm opening against 270 mm needed	R5 met
Cycle	5.2 min; 69 pots per 6 h	R4 met
Wall evenness	±1.40 mm as cast; ±0.43 mm finished to templates	R2 at risk
Flow gauge	0.1 L is 1.69 mm; 2.3 % per °C near 25 °C	R6 met
Size and mass	Press 840 x 640 x 1,806 mm, 298 kg; rack 780 x 780 mm	R11 not met for the rack
Cost	Press \$845, QC rack \$81, total \$926	R10 (\$720) not met

The working force is still an assumption. Henry, Maley and Mehta (2013) formed round-bottom filters with a 2 t car jack ([IJSLE 8 \(1\)](#)), which suggests the real force may be well below 5 to 10 t; the frame is sized for the full 20 t jack either way.

■ Key design choices

Items 1 to 7 and the budget were decided by Amish on 2026-09-25: go with recommendation (PPR-DDR-001).

- **Jack below, pushing the female mold up.** A bottle jack must stand upright, so it sits on the base and lifts the platen, and the male mold hangs from the top beam.
- **Male mold lift by hand crank with a load pin.** A standard jack gives only 150 mm of stroke, so the crank provides the other 200 mm of opening. The TRL 2 pair of 30 mm pins fails in bending by calculation; the model uses one 60 mm alloy-steel pin, and the choice between one pin with an interlock and two pins is proposed, awaiting Amish (PPR-DDR-001 item 13).
- **Molds cast in aluminum from printed patterns.** Cast as shells to keep them under 25 kg, with the cavity finished by hand to printed templates. The fiber-reinforced concrete backed variant stays documented as a low-cost alternative; building it is TRL 4 work and on hold. Scrap alloy must be lead-free (proposed, item 15).
- **Welded frame.** A fully welded frame is about 147 kg in one piece; bolting the four upright joints (4 x M20 8.8 each, 75 MPa at 294 kN) keeps every part under 40 kg and is proposed as the bolted variant (item 12).
- **Demold by sliding out and tilting the carriage.**
- **Manual QC rack with printed T-gauge.** The rack is 2 x 2 stations; a load-cell logger stays a later option.
- **Default acceptance band 1.0 to 2.5 L/h in the first hour, corrected to 25 °C.** Each factory may set its own band.
- **Budget \$720.** The priced BOM is \$926; options are in PPR-DDR-001 item 14, awaiting Amish.

■ Safety

Safety: PotPress is a 20 t hydraulic press with heavy moving parts, and the workshop around it handles silica-bearing clay dust and silver compounds. Treat each of these as a hazard at every stage.

- **Crushing and pinch points.** The gap between the molds, the platen and the uprights, and the carriage rails can crush fingers and hands. Press only with the side guards on and the front gate closed; the jack pump and release valve sit outside the guard; the carriage has an end stop so it cannot be pushed past the male mold.
- **Load pin and stored energy.** One 60 mm pin carries the whole press force. Never press unless the pin is fully home through both beam webs and the stem; the guard must include a pin-presence interlock or a blocking plate. Never exceed the jack rating, never add a cheater bar to the pump handle, and never adjust the jack's relief valve. Release pressure fully before opening the gate or pulling the pin.
- **Falling and tipping.** The molds weigh about 24 and 18 kg and the press about 298 kg with its center of mass about 0.8 m up. A push of about 740 N at 1 m tips it forward, so anchor the base to the floor, lift molds with two people and wear safety boots. The lead screw is self-locking, so the male mold holds its height if the handwheel is released; still keep hands out from under it.
- **Silica dust.** Dry clay and rice husk ash contain crystalline silica, which causes silicosis. Keep mixing and trimming wet, sweep damp, and wear a fitted P2 or N95 respirator for dry work.
- **Molten aluminum and alloy.** Casting is done by a foundry, not at the press. Scrap containing lead (free-machining alloys) must not be used for mold faces that touch a drinking-water product.
- **Silver compounds.** Silver nitrate is corrosive to skin and eyes and toxic to aquatic life; handle colloidal silver with gloves and eye protection and never pour it into waterways. Silver treatment is outside PotPress but happens in the same workshop.
- **Heat.** Kilns reach about 900 °C. Kilns are outside PotPress but share the site.
- **Water safety claims.** PotPress makes consistent pots; it does not certify them. A filter that passes the flow test is not proven to remove pathogens. Factories must verify bacteria removal by laboratory testing and follow the CMWG recommendations.

■ Open questions

- What pressing force gives a well-consolidated wall with a typical mix? Ask a partner factory; the 2 t figure above suggests it may be low.
- Can hand finishing to printed templates reach ± 0.3 mm on a 400 mm casting (R2, R7)?
- R3 deflection target, R11 rack area, frame joints, load pin, cost gap and scrap specification: PPR-DDR-001 items 10 to 15, awaiting Amish.
- Handwheel height of about 1.8 m: a side crank through a bevel gear would be easier to reach.
- Confirm the demolding method with potters; check whether a liner leaves marks that affect flow.
- Choose the first partner factory or organization (left open under the portfolio rule).

Concept media: [blueprint sheet](#), [interactive 3D model](#). General arrangement: [PPR-DWG-001](#).